

Evaluating Limits by Factoring and Rationalizing

Directions:

- Every limit below gives the indeterminate form $\frac{0}{0}$ on direct substitution.
- Rewrite the expression first — factor and cancel, or multiply by the conjugate — and only then substitute.
- Give exact values (fractions, not decimals). Write the limit notation on every line of work, not just at the end.

1. The table lists values of $f(x) = \frac{x^2 - 16}{x - 4}$ near $x = 4$.

x	3.9	3.99	4.01	4.1
$f(x)$	7.9	7.99	8.01	8.1

Predict $\lim_{x \rightarrow 4} f(x)$ from the table, then confirm the prediction by factoring the numerator and cancelling.

Answer: _____

2. Evaluate $\lim_{x \rightarrow -2} \frac{x^2 + 5x + 6}{x + 2}$

Answer: _____

3. Evaluate $\lim_{x \rightarrow 0} \frac{\sqrt{x+9} - 3}{x}$

Answer: _____

4. Evaluate $\lim_{x \rightarrow 3} \frac{x^2 - x - 6}{x^2 - 9}$

Answer: _____

5. Evaluate $\lim_{x \rightarrow 4} \frac{\sqrt{x} - 2}{x - 4}$

Answer: _____

6. The quotient below is the average rate of change of $y = x^3$ from $x = 2$ to $x = 2 + h$, so its limit is the instantaneous rate of change at $x = 2$. Expand the cube, cancel the h , and evaluate

$$\lim_{h \rightarrow 0} \frac{(2+h)^3 - 8}{h}$$

Answer: _____

7. The table lists values of $g(x) = \frac{x^3 - 8}{x^2 - 4}$ near $x = 2$, rounded to four decimal places.

x	1.9	1.99	2.01	2.1
$g(x)$	2.9256	2.9925	3.0075	3.0756

(a) Factor both the numerator and the denominator, cancel, and give the exact value of $\lim_{x \rightarrow 2} g(x)$: _____

(b) Explain why the four table entries on their own do not prove the value you found in part (a).

8. Challenge. The radical is in the denominator this time. Evaluate

$$\lim_{x \rightarrow 1} \frac{x - 1}{\sqrt{x + 3} - 2}$$

Answer: _____